

Lab 5 report

1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?

Raw audio is a long 1D signal where useful speech structure (formants and phoneme energy bands) is hard for a small KWS model to learn directly. Converting to spectrogram/micro/MFCC-style features gives a compact time-frequency fingerprint that captures which frequency bands are active over time, improves robustness to loudness/noise/phase variation, and reduces input size for TinyML (the notebook flattens inputs to 1x1960).

2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?

The representative dataset is used for calibration during conversion. In the notebook, `representative_dataset_gen()` yields real preprocessed feature samples from `audio_processor.get_data(...)`. TFLite uses these samples to estimate tensor min/max ranges and compute INT8 scale/zero-point values. Without representative data, ranges can be poorly estimated, causing clipping/saturation and significant accuracy loss in the quantized model.

3. What does `audio_processor.get_data()` do? Refer to `input_data.py`.

`audio_processor.get_data()` builds feature batches and labels from a selected partition (training/validation/testing). It selects samples, applies time shift, mixes background noise and silence handling, scales foreground volume, runs the configured preprocessing pipeline (average/mfcc/micro), flattens outputs to `fingerprint_size` vectors, and returns (data, labels).

4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array.

What are `g_model` and `g_model_len`, and why do they need to be copied correctly into the Arduino source file?

`g_model` is the byte array containing the full .tflite FlatBuffer model. `g_model_len` is the exact byte length of that array (in this notebook output, 19104). TFLite Micro on Arduino reads this embedded array as the model binary. If bytes or length are incorrect/mismatched, model parsing can fail or load corrupted/truncated data, causing initialization or inference errors.

Result:

```
Heard unknown (222) @104272ms
Heard unknown (223) @105792ms
Heard unknown (217) @107504ms
Heard unknown (246) @109024ms
Heard unknown (208) @110544ms
Heard yes (203) @112608ms
```

AI disclosure:

Codex is used for code check and report writing style and logic check